

Phishing Awareness Analysis

Cyber Security Internship – Project 3

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Technology Used: Python

1. Introduction

Phishing is a common cybersecurity threat where attackers use fake or deceptive emails and messages to trick users. This project focuses on analyzing sample messages and identifying suspicious indicators that may represent a phishing attempt.

2. Objective

The main objective of this project is to analyze sample emails or messages and identify possible phishing attempts. The project focuses on:

- Identifying suspicious links or keywords
- Listing red flags found in phishing messages
- Explaining why the message may be unsafe

3. Technology Used

Programming Language: Python

Python is used to analyze the message entered by the user and identify common phishing indicators.

4. Project Features

- Accepts an email or message as input.
- Detects suspicious keywords.
- Detects suspicious links.
- Displays possible phishing warnings.
- Lists the detected red flags.
- Explains why the message may be unsafe.

5. Working of the Project

- The user enters an email or message.
- The program checks the message for suspicious keywords.

- The program checks for suspicious link patterns.
- If suspicious indicators are found, the program displays *Possible Phishing Message*.
- The detected red flags are displayed.
- The program explains why the message may be unsafe.
- If no common indicators are found, the program displays *No common phishing indicators found*.

6. Suspicious Keywords Checked

The program checks for keywords such as:

Urgent	Verify	Winner
Prize	Click here	Password
Account blocked	Limited time	Claim now

7. Suspicious Links Checked

The program checks for common suspicious link patterns such as:

- http://
- bit.ly
- tinyurl.com

8. Sample Test Case – Phishing Message

Input:

URGENT! Your account is blocked. Verify your password now by clicking here: http://example.com

Output:

Possible Phishing Message

Red Flags Found:

- Suspicious keyword: urgent
- Suspicious keyword: password
- Suspicious keyword: account blocked
- Suspicious keyword: click here
- Suspicious link found: http://

Reason:

The message contains suspicious keywords, urgency, a password verification request, and a link. Therefore, the user should avoid clicking the link or sharing personal information.

9. Sample Test Case – Normal Message

Input:

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Hello, your meeting is scheduled for tomorrow at 10 AM.  
Please be on time.
```

Output:

```
No common phishing indicators found.
```

Reason:

The message does not contain the suspicious keywords or link patterns checked by the program.

10. Key Learning

This project helped me understand phishing indicators, threat analysis, and cybersecurity awareness. I also learned how simple Python logic can be used to identify common warning signs in messages.

11. Conclusion

Phishing awareness is an important part of cybersecurity. This project demonstrates a simple method for analyzing messages and identifying common phishing indicators. The project guidelines also suggest practical ideas such as creating a Red Flag Checklist and understanding how URL shorteners can hide malicious domains.

12. Future Enhancements

- Add more phishing keywords.
- Improve suspicious URL detection.
- Add sender email analysis.
- Create a Red Flag Checklist.
- Detect suspicious shortened URLs.
- Add a graphical user interface.